

MT 202 : Applied Quantum Mechanics

Course Code : MT 202

Course Name : Applied Quantum Mechanics

L-P-T-C: : 3-0-0-3

Intended for: B.Tech. (MSE)

Prerequisites: None

Mutual Exclusion: None

Approval : 57th BoA

Course Contents

- **Module 1:** Review of classical mechanics and its failure in understanding the behaviour of microscopic particles, Waves and particles (double-slit experiments), Introduction to the basic ideas of quantum mechanics, The postulates of quantum mechanics, Eigen-states and eigen-values of position and momentum operators, Uncertainty principle. Observables and their operators. [10 hours]
- **Module 2:** Time independent and time dependent Schrodinger's equations, The fundamental properties of Schrodinger's equations, One-dimensional square well potential, The electron gas in a 3-dimensional box, Tunneling, The one-dimensional Harmonic oscillator, Vibration of the nuclei of a diatomic molecule, Vibration of the nuclei in a crystal. [10 hours]
- **Module 3:** General properties of angular momentum, Commutation relations characteristic of angular momentum, General theory of angular momentum, Electron spin, Particle in a central potential, The hydrogen atom, The hydrogen atom placed in a uniform magnetic field, Paramagnetism and diamagnetism, The Zeeman effect. [10 hours]
- **Module 4:** Stationary Perturbation theory: non degenerate and degenerate cases, A one-dimensional harmonic oscillator subjected to a perturbing potential in x , x^2 , and x^3 , The variational method, Energy bands of electrons in solids: a simple model. [6 hours]
- **Module 5:** Quantum confinement in nanostructures, and important example problems like optical absorption in semiconductors, Experimental realization of quantum mechanics; STM, LIGO Interferometer, SQUID, nano-optomechanics, optical cavities, quantum sensor; atomic/ion trap, atomic clock etc. Brief introduction about the qubit, logic gate, Bell's inequality, entanglement, quantum interference. [6 hours]

Text books and References:

1. Claude Cohen-Tannoudji, Bernard Diu, and Franck Laloe, Quantum Mechanics: Vol. 1 & 2, 2nd Edition, Wiley-Vch, 2020.
2. R. Shankar, Principles of Quantum Mechanics, 2nd Edition, Plenum Press, 1994.